

Course: Biological Science for Engineers (BE24101)

Origin of Life



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Biological Science for Engineers

M1: Introduction to Biological Sciences

- Overview and importance of biology in engineering
- Origin of Life
- Cell Theory, Cell Structure and Function

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Module -I

Origin of Life



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What is Life ?

Life is an Adventure.....Dare it.

Life is a Love.....Enjoy it.

Life is a Tragedy.....Face it.

Life is a Struggle.....Fight it.

Life is a Promise.....Fulfill it.

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What is Life ?

Life is like a camera...
Focus on what's important,
Capture the good times,
Develop from the negatives,
 And if things don't work out,
Take another shot.

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What is Life ?

- It's a characteristic that distinguishes objects that have signaling and self sustaining processes from those that do not have because such functions have ceased or because they lack such functions.



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Characteristics of Living things

- **Survivability**
- **Adaptations**
- **Responsiveness**
- **Genetic composition**
- **Mutation**
- **Evolution**
- **Genetic variations**
- **Cellular composition**
- **Metabolism**
- **Movement**
- **Reproduction**
- **Development**
- **Growth**
- **Respiration**



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Life's unanswered questions...

- How does life began on the Earth and evolve?
- Does life exist elsewhere in the universe?
- Relationship to early Earth conditions?
- Future of life on Earth and beyond?



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Life's unanswered questions...

- How does life began on the Earth and evolve?
- Does life exist elsewhere in the universe?
- Relationship to early Earth conditions?
- Future of life on Earth and beyond?
- Limits of life on the Earth?

Knowlegde



Speculations



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Life's unanswered questions...

CHANDRAYAAN-3

**SUCCESSFULLY
LANDS ON MOON**



India's third lunar mission that took off from the Satish Dhawan Space Center on July 14, 2023, at 2:35 pm.

Mission Objectives of Chandrayaan-3:

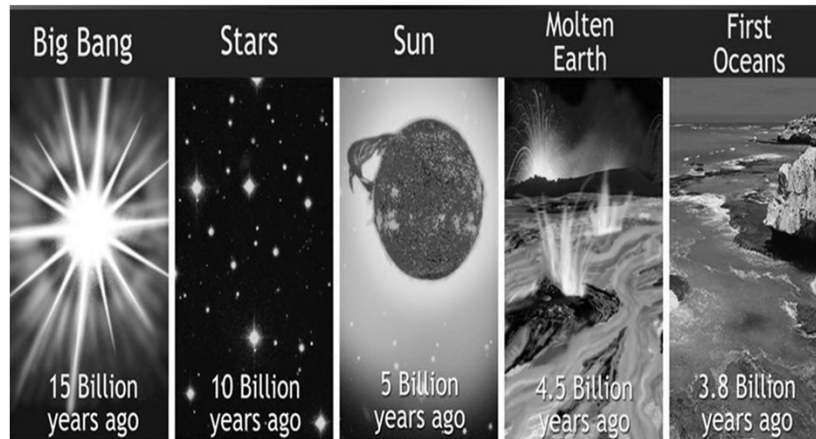
- To demonstrate Safe and Soft Landing on Lunar Surface
- To demonstrate Rover roving on the moon and
- To conduct in-situ scientific experiments.

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Module - I

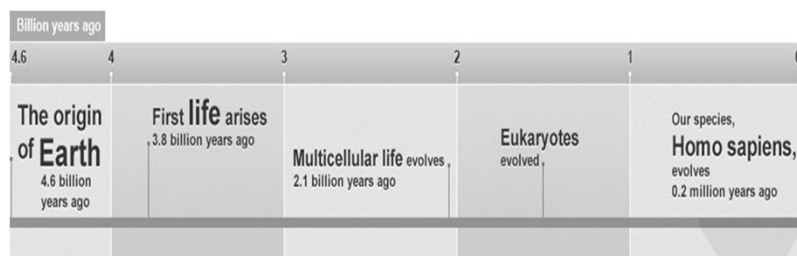
Timeline of Earth



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Module - I

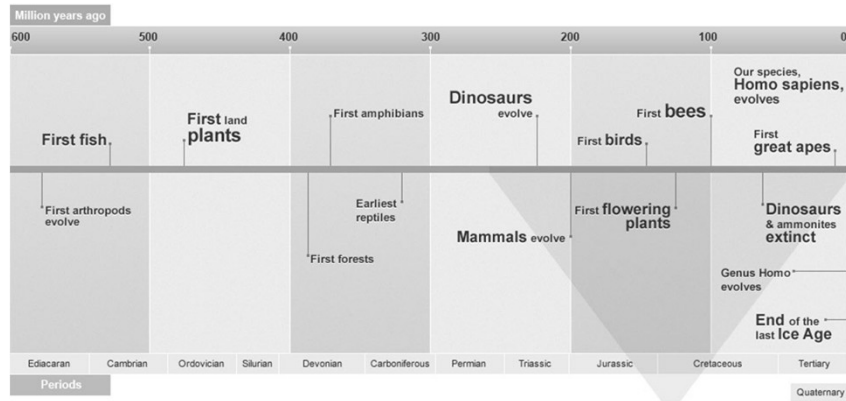
Timeline of life on Earth



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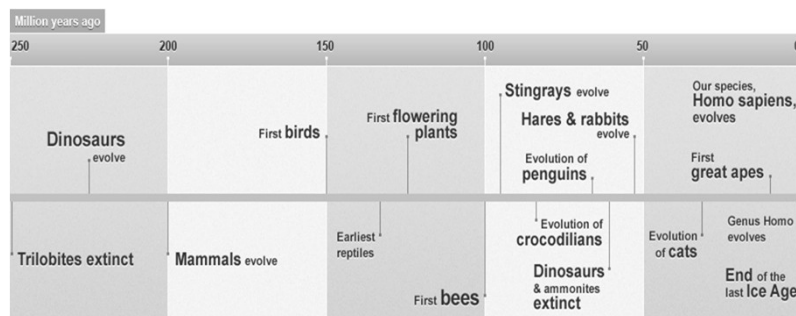
Timeline of life on Earth



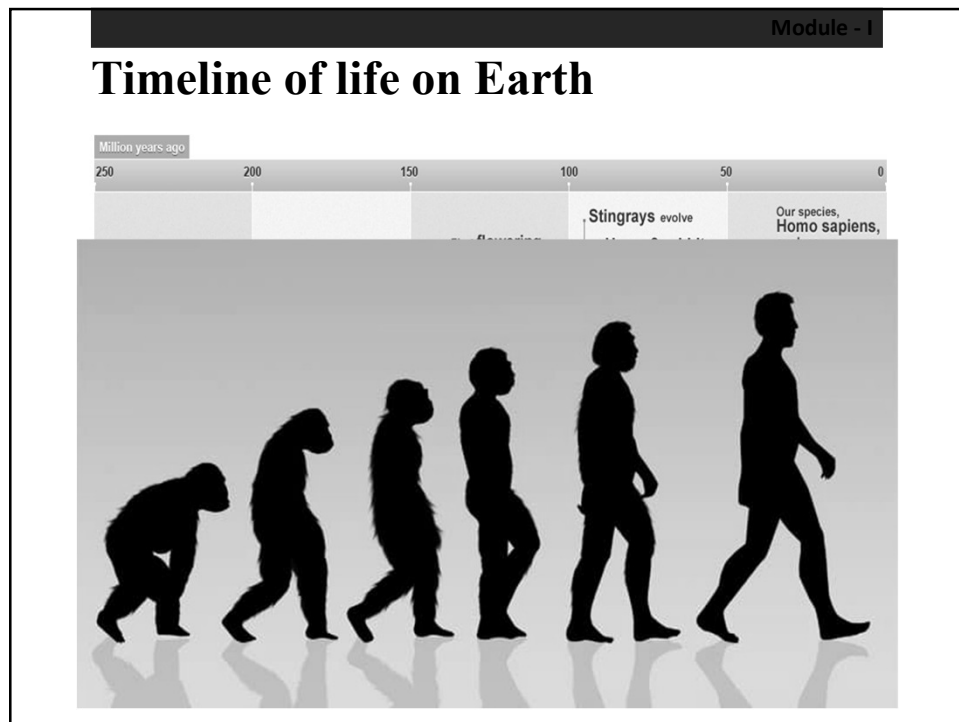
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Module - I

Timeline of life on Earth



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Module - I

First programmable computer

The Z1 was **created** by German Konrad Zuse in his parents' living room between 1936 and 1938. It is considered to be the **first** electromechanical binary programmable **computer**, and the **first** really functional modern **computer**.

Jun 30, 2019

<https://www.computerhope.com> » is...

When was the first computer invented? - Computer Hope

1835

One of the first electronic devices was invented around **1835** by an American scientist called Joseph Henry. He developed a remote switch called a relay, which worked magnetism and currents. It was part of a telegraph and was later used in a telephone. Sep 25, 2018

Who invented the first electric circuit?

Alessandro Volta

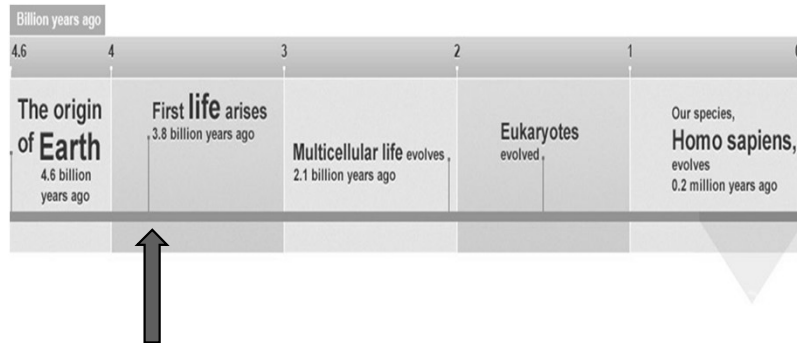
The first electric circuit was invented by **Alessandro Volta** in 1800. He discovered he could produce a steady flow of electricity using bowls of salt solution connected by metal strips.

<https://science.jrank.org> » pages » El...

Electric Circuit - Current, Electrical, Resistance, and Series - JRank ...

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Timeline of life on Earth

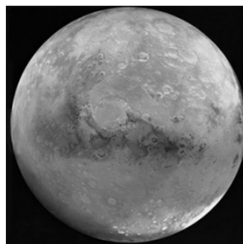


- ✓ Non conductive environment
- ✓ Extremophilic conditions

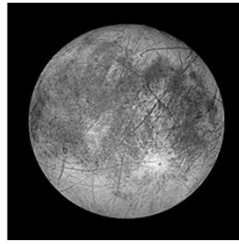
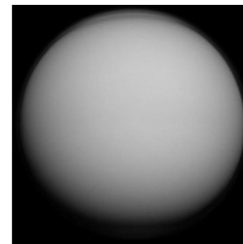
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Life's unanswered questions...

- Studies of life in extreme environments on Earth have led us to focus on some prime places to look for life.

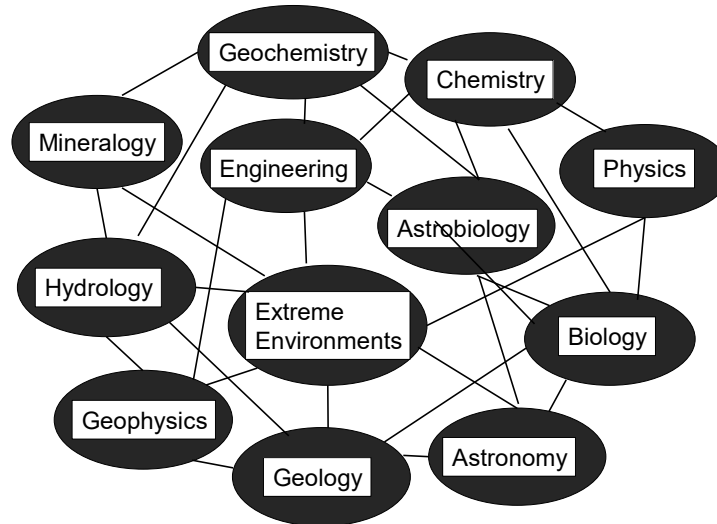


Mars

Europa
(moon of Jupiter)Titan
(moon of Saturn)

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Intigration of different Sciences



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Module - I

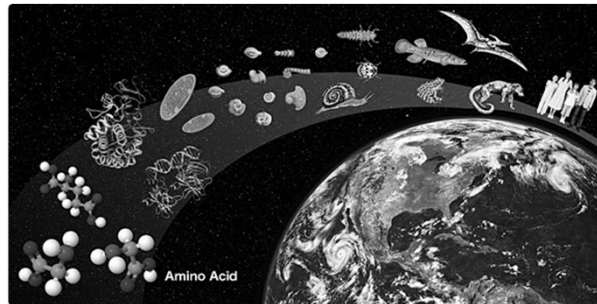
Origin of life

- The origin of life was a unique event in the history of universe.
- Universe is ~15-20 billion years old and comprised of huge clusters of galaxies that contain stars and clouds of gas & dust. The **Big Bang** resulted in the formation of universe. H_2 and He formed.
- The Earth was supposed to have been formed 4.5 billion years back with no atmosphere – Early Earth.
- Water vapour, CH_4 , CO_2 & NH_3 released from molten mass. Then Ozone was formed.
- Early Earth was a molten mass and gradually it cooled and at the same time water vapours fell as rain, to fill all the depressions and form oceans.
- Life appeared after the formation of Earth i.e. ~3.8 billion years back.

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Origin of life on the Earth

Several attempts have been made from time to time to explain the origin of life on Earth and as a result there are several theories which have their own explanations.



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Origin of life on the Earth

- 1. Theory of Special Creation**
- 2. Spontaneous Generation Theory**
- 3. Theory of Catastrophism**
- 4. Cosmozoic Theory**
- 5. Theory of Chemical Evolution**

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Module - I

Theory of Special Creation



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Module - I

Theory of Special Creation

Hindu concept

Lord Brahma created the living world in one stroke.



Cristian & Islam concept

God created this universe, plants, animals & human beings in about 6 days.



Sikh concept

Guru Nanak.....Guru Gobind

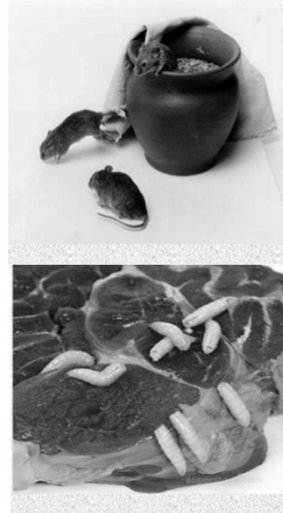
Its purely a religious concept

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Module - I

Theory of Spontaneous generation

- Sudden and spontaneous from non living things
- Aristotle – Dead leaves falling in pond gave rise to Fishes and falling on soil gave rise to Worms and Insects.
- Egyptians – mud of Nile river – many life forms.
- Von Helmont- Soup experiment



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Module - I

Theory of Spontaneous generation

- Later disapproved by Louis Pasteur (1864)

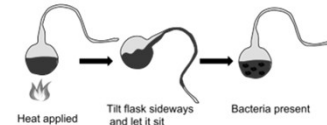
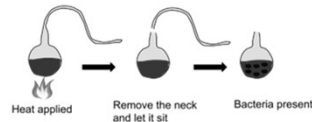
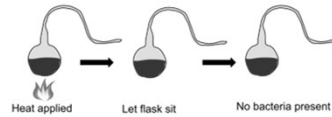


Louis Pasteur (1822-1895), French Chemist, well known for developing the principles of vaccination and pasteurization

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Theory of Spontaneous generation

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Swan Neck Experiment



Louis Pasteur (1822-1895), French Chemist, well known for developing the principles of vaccination and pasteurization

'Life comes from pre-existing life'

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Theory of Catastrophism

- Modification of theory of special creation.
- It states that there has been several creations of life by God, each produced by a catastrophe.
- Each catastrophe completely destroyed the existing life on the Earth and a different form of life was recreated.



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Cosmozoic / Theory of Panspermia

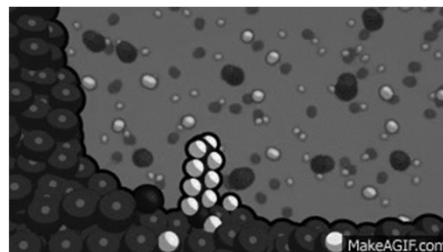
- The Earth was seeded with primitive life forms from other planets in the form of spores came along with meteorites.
- The spore has to be very small, radiation resistant and has to travel under vacuum.
- Richter (1865) and Arrhenius (1908) supported this theory.



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Theory of Chemical Evolution

- Materialistic or Physico-chemical theory.
- The origin of life is the result of slow and gradual process of chemical evolution that probably occurred 3.8 billion years ago.



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Theory of Chemical Evolution

- The origin of life is the result of slow and gradual process of chemical evolution that probably occurred 3.8 billion years ago.



Alexander Oparin (1894-1980)
Russian Biochemist



J.B.S. Haldane (1892-1964)
British Biologist

Theory of Chemical Evolution

1. Spontaneous generation of life is not possible.
2. Earth's surface & atmosphere during the first billion years of existence were different from today.
3. The primitive Earth's atmosphere was a reducing type of atmosphere.
4. The first life arose from a collection of chemical substances through a series of chemical reactions.
5. Solar radiations, heat radiated from Earth and lightning must have been the chief energy source for these reactions.

Module - I

Theory of Chemical Evolution

- Oparin and Haldane proposed a hypothesis in 1950-

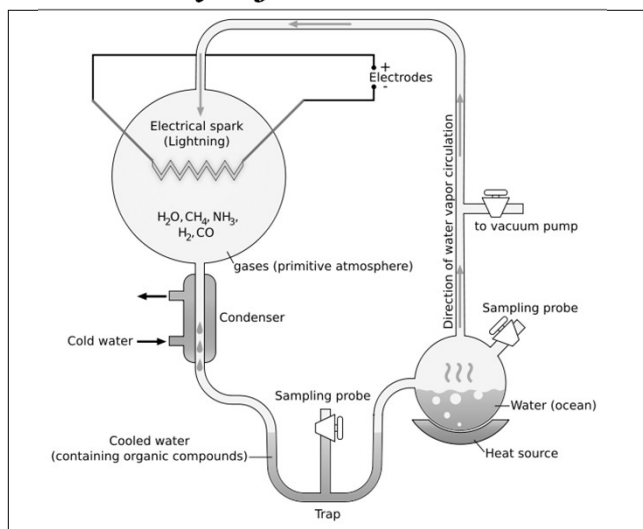
“Conditions on the primitive Earth favored chemical reactions that synthesized organic compounds from inorganic precursors”



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Module - I

Theory of Chemical Evolution



Miller's experiment (1953)



Stanley L. Miller
American Chemist



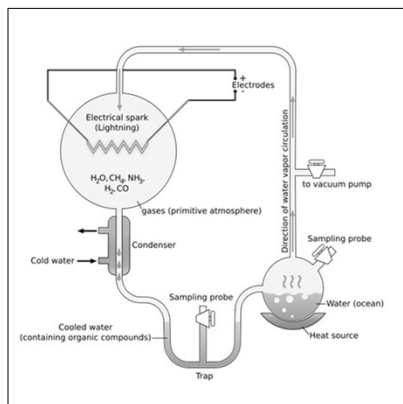
Harold C. Urey
American Physical
Chemist

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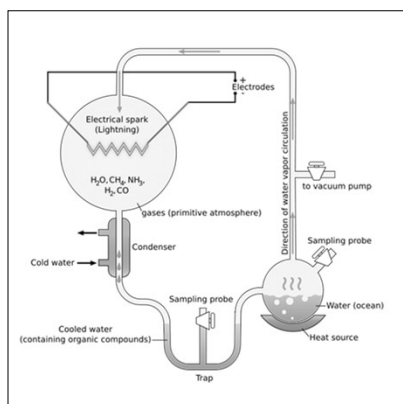
Theory of Chemical Evolution

**Formaldehyde****Acetaldehyde****Glycine****Alanine****Glutamic acid****Aspartic acid****Succinic acid****Formic acid**

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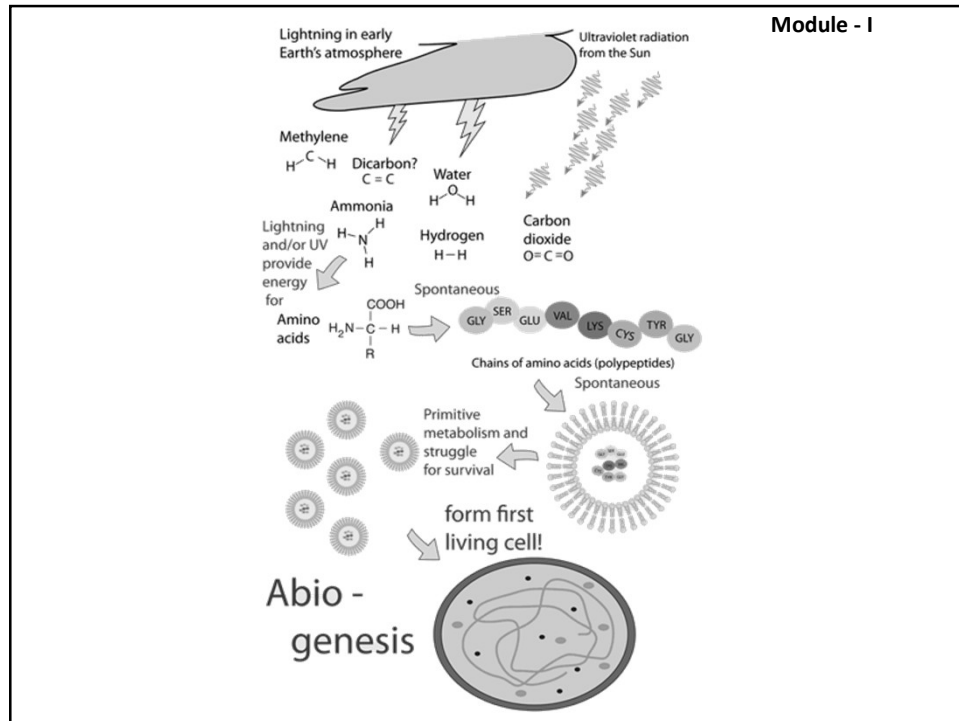
Module - I

Theory of Chemical Evolution

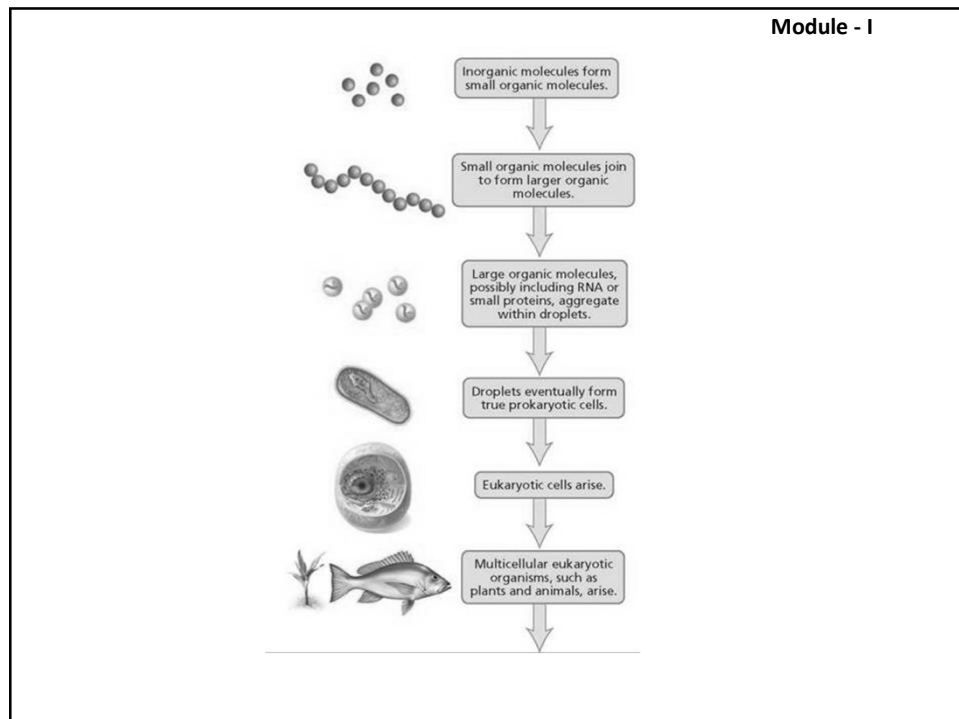


1. 10-15% carbon was now in the organic compounds.
2. 2% of the carbon had formed some amino acids.
3. They showed that organic compounds could be made easily under the conditions that scientists believed to be present on the early Earth.

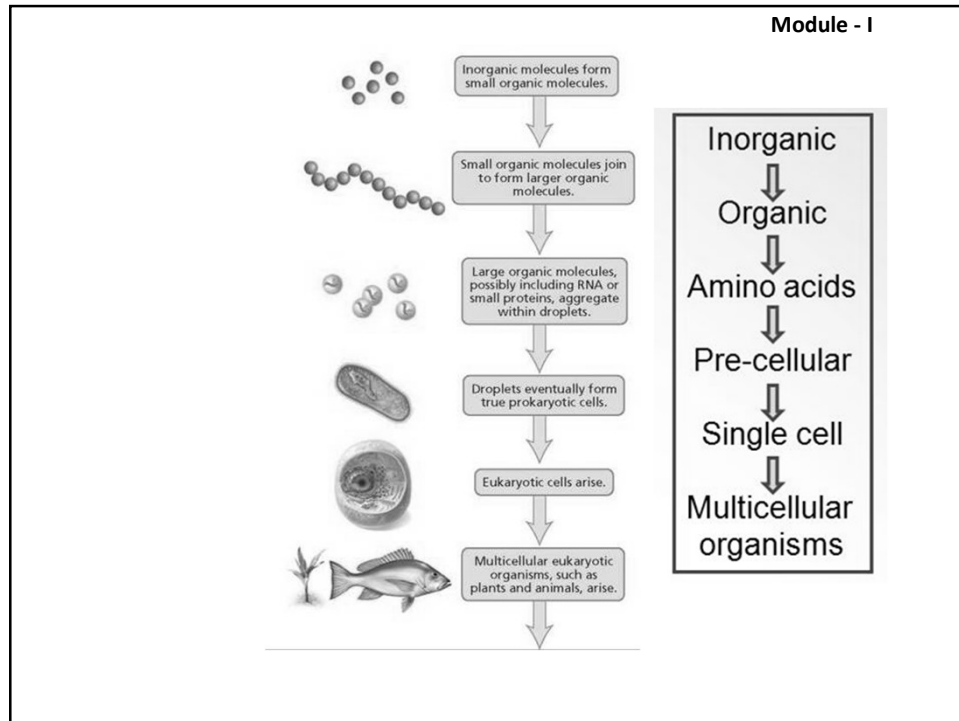
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THANK YOU

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